Pedestrian Perception of Autonomous Vehicles with External Interacting Features

Christopher R. Hudson[1], Shuchisnigdha Deb[1(✉)], Daniel W. Carruth[1], John McGinley[1], and Darren Frey[2]

[1] Center for Advanced Vehicular Systems, Mississippi State University, Starkville, USA
{chudson, deb, dwc2, jam}@cavs.msstate.edu
[2] Industrial and Systems Engineering, Mississippi State University, Starkville, USA
dafl90@msstate.edu

Abstract. The increasing number of autonomous vehicles has raised questions regarding pedestrian interaction with autonomous vehicles. Researchers have studied external interfaces designed for vehicle operators and other road-users (e.g., pedestrians). Most past studies have considered the interaction between pedestrians and autonomous vehicles with no visible operator. However, pedestrian-autonomous vehicle interaction may be complicated when there is a human sitting in the conventional driver's seat of an autonomous vehicle. Such a scenario may cause some pedestrians to look to the passenger for cues when they should be looking for cues from the vehicle. The objective of the current study was to investigate pedestrians' perspective of autonomous vehicles based on the interaction effect between passenger status and external features on the vehicle. Sixteen pedestrians completed a VR experiment. The results provided important insight into the important question of pedestrian-autonomous vehicle interaction when passengers are present in the driver seat of the vehicle.

Keywords: Autonomous vehicles · Pedestrian-autonomous vehicle interaction Virtual reality · Communication interface design

1 Introduction

An increasing number of autonomous vehicles are being integrated into every day traffic patterns. New advancements in self-driving vehicles are being rapidly developed and tested on roadways by major motor vehicle manufacturers and technology companies. TESLA has deployed semi-autonomous vehicle control systems to many of its vehicles with many more systems expected to follow suit. While many current systems are restricted to advanced driver assist features or basic autonomous behaviors such as lane keeping and collision avoidance, the long-term goal of vehicle manufacturers is to implement fully autonomous vehicles that minimize or eliminate direct human control. More than 20 states have authorized operation of autonomous vehicles since 2011 [1]. While these systems are experimental and have been involved in accidents [2, 3], as vehicles acquire more autonomy and operate in close proximity to other vulnerable

© Springer International Publishing AG, part of Springer Nature 2019
I. L. Nunes (Ed.): AHFE 2018, AISC 781, pp. 33–39, 2019.
https://doi.org/10.1007/978-3-319-94334-3_5

road users, the autonomous vehicles must successfully navigate the unspoken social etiquette that exists between passengers in vehicle and pedestrians.

The number of accidents at intersections involving pedestrians has increased over time. In 2015, there were 5376 pedestrian fatalities reported by the National Highway Traffic Safety Administration (NHTSA) [4]. This was a 9.5% increase over 2014 and resulted in the highest fatality rate since 1996. A total of 967 fatalities occurred at intersections with 76% of the fatalities occurring in urban areas. Autonomous vehicle technology promises to make roads safer for pedestrians and other vulnerable road users. However, pedestrians will need to interact safely with autonomous vehicles. Today, when a pedestrian interacts with a vehicle and its driver, the pedestrian and the driver may be able to make eye contact and exchange non-verbal cues or signals that help the pedestrian to know whether it is safe to cross. When a pedestrian meets a fully autonomous vehicle, pedestrians are unable to make "eye contact" with the operator as suggested by the NHSTA. One could argue that if no-one is in the vehicle, the pedestrian will know that a vehicle is autonomous and rely on other cues such as visibly slowing of the vehicle or displays on the autonomous vehicle. However, another scenario exists when a passenger in the autonomous vehicle is occupying the seat of the traditional vehicle operator. In this scenario, a pedestrian may be misled by the behavior of the apparent operator of the vehicle. For example, the passenger of the vehicle may be looking down and away from the road indicating that they are not aware of the pedestrian and the situation is not safe. However, the autonomous vehicle is aware of the pedestrian and will allow the pedestrian to safely cross the street. This scenario raises questions about how pedestrians will interpret conflicting signals from passengers and from the vehicle. We will explore pedestrians' responses to different features and autonomous vehicle behaviors to advance our understanding of autonomous vehicle and pedestrian interaction.

This study extends previous research in autonomous vehicle and pedestrian interaction at crosswalks in order to understand the interaction in the case where a passenger in the fully autonomous vehicle is seated in the traditional operator's seat and may be interpreted as being in control of the vehicle. We selected visual and audio features from previous survey and real-world studies on pedestrian crossing with an autonomous vehicle [5–7]. A vehicle passenger seated in the traditional operator's seat was added to the simulated vehicles in certain conditions. The passenger's status was one of three levels: no passenger, a passenger with head up and apparently paying attention, and a passenger with head down and distracted. Across all three passenger conditions, the vehicle had seven different external features: no feature, green *walk* in text, white walking silhouette, red upraised hand, stop sign, music, and verbal message saying *safe to cross*. For human safety, real world experimentation to investigate human-autonomous vehicle interaction cannot be approved by research ethics. Therefore, to create a controlled environment and expose participants to a fully autonomous vehicle without any deception, this study was performed in an ambulatory virtual reality environment [8].

2 Method

2.1 Participants

The study was approved by the Institutional Review Board at Mississippi State University. Sixteen participants (9 male and 7 female), ranging in age from 19 to 34 years old, were recruited from the Starkville, Mississippi area to participate in a virtual reality experiment. The participants were selected based on their language (English speakers), ability to walk at normal pace and gait, and normal or corrected-to-normal color vision.

2.2 Apparatus and Setting

Traffic environments were created using Unity 2017.3 and experienced with HTC Vive headset. Two lighthouse sensors of the HTC Vive tracked participants' head movements during their crossing across the virtual crosswalk. The lighthouse sensors were placed approximately 8 m apart, facing each other at opposite ends. The area tracked by the Vive was approximately 4 by 7 m that included a 5.5-meter-long crosswalk and sidewalks on both sides.

Artist models of the Google autonomous car, other cars and trucks, and a human were purchased from the Turbosquid website (www.turbosquid.com). Animations for the human models were created using Autodesk 3D Studio Max 2018. The attentive passenger's head was up and scanning left and right to give an impression that he was aware and observing the environment around the vehicle. The distracted passenger's head was looking down and his hands were moving giving the impression that he was working on a tablet.

2.3 Experimental Protocol

After the participants arrived at the lab, they read and signed an informed consent form. Then the participants completed a Simulation Sickness Questionnaire (SSQ, Kennedy et al., 1993) to establish a baseline score and to confirm their ability to participate in the study. After completing the SSQ survey, the participants put on the VR headset to experience and get used to moving and acting in virtual reality. The initial environment helps move participants past their initial excitement about VR and teach them to receive and respond to instruction within the virtual world. After their first exposure to virtual reality, the participants completed a second SSQ survey to assess any immediate discomfort from virtual reality. After this second SSQ, the participants were immersed in a virtual traffic environment with a two-lane four-way intersection and with two-way stop signs. They were positioned on a sidewalk facing a crosswalk, with the intersection and a two-way stop sign to their left. The vehicles were running only on the two lanes perpendicular to the participant's position and no stop sign was present in those lanes.

The first three trials provided practice crossing the street in the virtual scenario with three vehicles passing by from both directions of the two-way lane. The participants crossed the road when they felt it was safe to do so. Once the participants felt

comfortable with the three practice crossings, they initiated the experimental trials by following instructions presented within the virtual world. During the experimental trials, the six vehicles (3 on each lane, as in the practice trial) and a fully autonomous car approached the crosswalk. The autonomous vehicle approached from the right and far lane. The autonomous car (trial car) was positioned in such a way within the scenario that even though all the other cars had passed by, the gap between the trial car and its lead vehicle was not sufficient for the participant to safely cross in front of the trial car if it did not stop. The car was equipped with one of seven levels of features: one no feature, two audible features, and four visual features. A display was installed on the front of the car to show no feature, green *walk* in text, white walking silhouette, red upraised hand, or stop sign. Sound cues provided audible features by playing music or a verbal message saying *safe to cross*. There were three different passenger statuses: no passenger, attentive passenger looking at the surrounding, and distracted passenger working on a tablet. All vehicles were moving at 30 mph at the start of the trial. The driverless car began to decelerate at around 175 feet away and to activate display features at around 210 feet away from the crosswalk. However, the car did not stop when the visual features were the upraised hand or the stop sign. Participants crossed the road for twenty-one experimental trials with 3 × 7 feature combinations. Figure 1 shows some of the features installed in cars with different passenger statuses. At the completion of the experiment, the participants responded to a five-item demographic survey and filled out rating scales for each of the features based on the different passenger statuses. Each of the participants were compensated with $10 for their participation (Fig. 1).

Fig. 1. Examples for experimental vehicles equipped with features

3 Results

3.1 Effect of Feature Type and Passenger Status

An ANOVA was conducted for the feature ratings considering both factors: feature type and passenger status. The analysis found no interaction effect between these factors with respect to the feature rating. However, significant main effects were observed for feature type [$F(6, 12) = 132.157$, $p = < 0.0001$] and passenger status [$F(2, 12) = 12.247$, $p < 0.01$].

Descriptive statistics for average feature ratings based on each of the driver statuses and each of the features types are presented in the following figures. The verbal message was rated as the most preferred audible feature and *walk* in text was rated the most preferred visual feature. For the features that implied "*do not cross*", the stop sign was more preferred than the upraised hand. In the case of different passenger statuses, features installed in no passenger and attentive passenger vehicles were preferred as compared to the distracted passenger vehicle (Fig. 2).

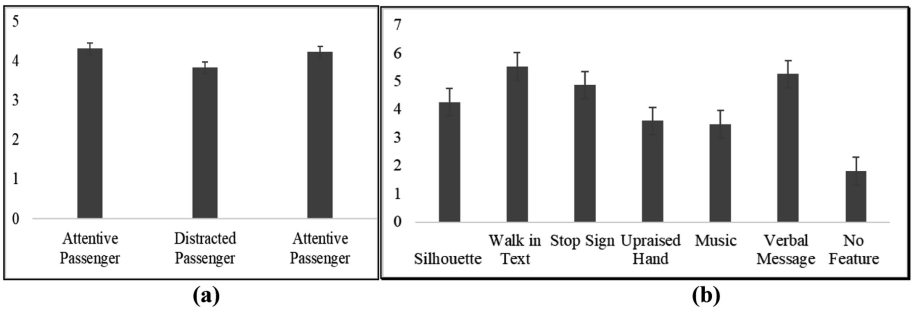

Fig. 2. Average feature ratings for different levels of (a) passenger statuses and (b) feature types.

3.2 Effect of Gender and Age

Another ANOVA was performed to observe demographic effect including data for gender (categorical variable) and age (continuous variable) along with the two factors. The results revealed that gender had a significant interaction effect with feature types to influence feature rating [$F(6, 12) = 3.571$, $p < 0.05$].

Average feature ratings for each of the feature types based on gender are shown in Fig. 3. The statistics shows that for no feature condition and for all of the display conditions where the car stopped at the intersection, males and females provided very similar ratings. However, for the display conditions where the car did not stop at the intersection, male participants rated both features with significant lower ratings as compared to the female participants. Both gender groups rated music with significantly lower scores as an option to feel comfortable crossing road as compared to the other audible feature.

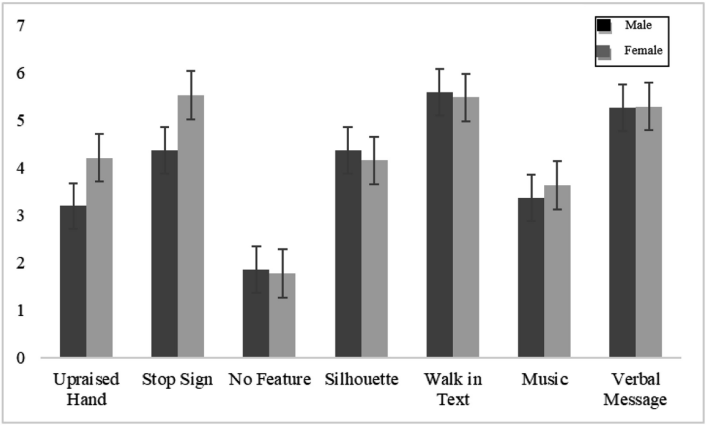

Fig. 3. Average feature ratings for different feature types based on gender

4 Discussion

Pedestrians' self-reported ratings were found to be influenced by both passenger status and feature type. In addition, there was no significant difference between feature ratings when the vehicle had no visible passenger and when the passenger of the vehicle appeared to be attentive. Pedestrians appear to feel comfortable communicating with an autonomous vehicle without any passenger as well as with a passenger who is paying attention. Therefore, it can be said that external features are useful replacements for cues that would normally be available from an attentive passenger. However, an autonomous vehicle with a distracted human passenger made the pedestrians' uncomfortable and cautious about the vehicle's intended action. This suggests that participants are confused when the autonomous vehicle is signaling intention to the pedestrian while cues from the apparent passenger indicate a lack of awareness of the pedestrian. Feature type also significantly influenced participants' rating for features. Participants were very comfortable with the walk, silhouette, and verbal message external features to confidently cross the road in front of an autonomous vehicle. Most of them also understood two negative signals, upraised hand and stop sign, as an indication for not to cross the road in front of that approaching autonomous vehicle. However, the upraised hand option was rated with significantly lower score than the stop sign option. In the case of the audible music, it might have interpreted by the participants as a negative and alarming signal to alert them not to cross the road and was thus rated lower as compared to the other audible feature. Lower scores on very familiar and standard image options like silhouette and upraised hand might indicate problems with visual presentation, which needs to be further investigated in the future research. While considering the effect of gender, it seemed like male participants were expecting the vehicles to stop all the time at the crosswalk and therefore, rated the features associated with failure to stop with lower ratings as compared to the female participants' ratings. The age range used in the study was not large enough (19–36 years) to show any significant effect of age.

5 Conclusion

The current study investigated participants' perspective toward an autonomous vehicle without any visible human passenger as well as with an attentive and a distracted passenger. The study allowed pedestrians to experience an autonomous vehicle in a simulated crosswalk and communicate with the vehicle. This research expanded Deb study [9] by including features that were recommended by most of the previous studies on this area. The consideration of different passenger statuses was a novel inclusion in this research.

The results came up with a short list of features for safe crossing conditions such as *Walk* in text, white walking silhouette, verbal message saying *safe to cross*, and stop sign for unsafe crossing condition. The authors plan to examine the effectiveness of these features on different type of autonomous vehicles (for example, olli, bus, etc.) with different positions and designs for display. The future research should also consider recruiting population with large age range to investigate age affect. Applied human factors researchers can use virtual reality experiments in user interface design, not limited to pedestrian-autonomous vehicle interaction.

References

1. National Highway Traffic Safety Administration: Traffic Safety Facts 2015 Data, February 2017. https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/812375
2. Matthew, D.: Tesla's fatal Autopilot crash is a reminder that we are still a long way from truly autonomous vehicles (2017). http://www.businessinsider.com/tesla-fatal-autopilot-accident-autonomous-cars-years-away-2017-9
3. Samuel, G.: Self-driving bus involved in crash less than two hours after Las Vegas launch (2018). https://www.theguardian.com/technology/2017/nov/09/self-driving-bus-crashes-two-hours-after-las-vegas-launch-truck-autonomous-vehicle
4. National Conference of State Legislation. http://www.ncsl.org/research/transportation/autonomous-vehicles-self-driving-vehicles-enacted-legislation.aspx
5. Fridman, L., Mehler, B., Xia, L., Yang, Y., Facusse, L.Y., Reimer, B.: To Walk or Not to Walk: Crowdsourced Assessment of External Vehicle-to-Pedestrian Displays. arXiv preprint arXiv:1707.02698 (2017)
6. Deb, S., Warner, B., Poudel, S., Bhandari, S.: Identification of external design preferences in autonomous vehicles. In: Proceedings of the 2016 Industrial and Systems Engineering Research Conference, Anaheim, California (2016)
7. Clamann, M., Aubert, M., Cummings, M.L.: Evaluation of Vehicle-to-Pedestrian Communication Displays for Autonomous Vehicles (No. 17–02119) (2017)
8. Deb, S., Carruth, D.W., Sween, R., Strawderman, L., Garrison, T.M.: Efficacy of virtual reality in pedestrian safety research. Appl. Ergon. **65**, 449–460 (2017)
9. Deb, S.: Pedestrians' receptivity toward fully autonomous vehicles. Doctoral dissertation, Mississippi State University (2017)